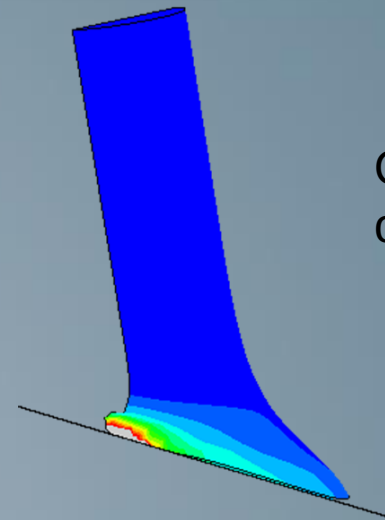


Thermo-Mechanical Material Models



3DEXPERIENCE



Oblique Impact
of a Copper Rod

Thermo-Mechanical Material Models

- ▶ How do we define 'thermomechanical' material model?
 - ▣ Thermal properties and mechanical properties together
- ▶ Are E and ν together with a CTE thermomechanical ?

- ▶ From Webster: *Adjective* ther·mo·mechanical
designed for or relating to the transformation of heat energy into mechanical work

- Of or pertaining to the variation of the mechanical properties of a material with temperature

Thermo-Mechanical Material Models

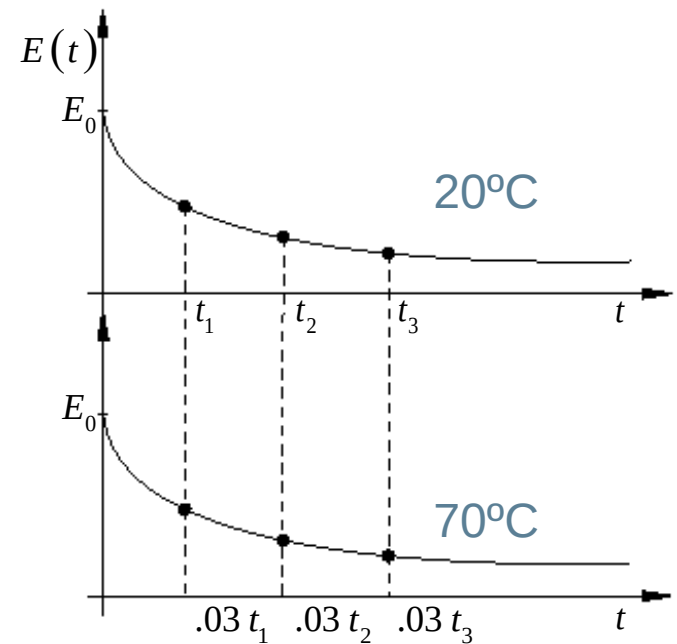
- ▶ For metals we give many nice examples:
 - Benchmark **1.6.4 Quenching of an infinite plate** (both)
 - Example Problem **1.3.16 Upsetting of a cylindrical billet** (both)
 - Example Problem **1.3.18 Inertia welding simulation** (Abaqus/Standard)
 - Example Problem **2.1.13 Oblique impact of a copper rod** (Abaqus/Explicit)

Thermo-Mechanical Material Models

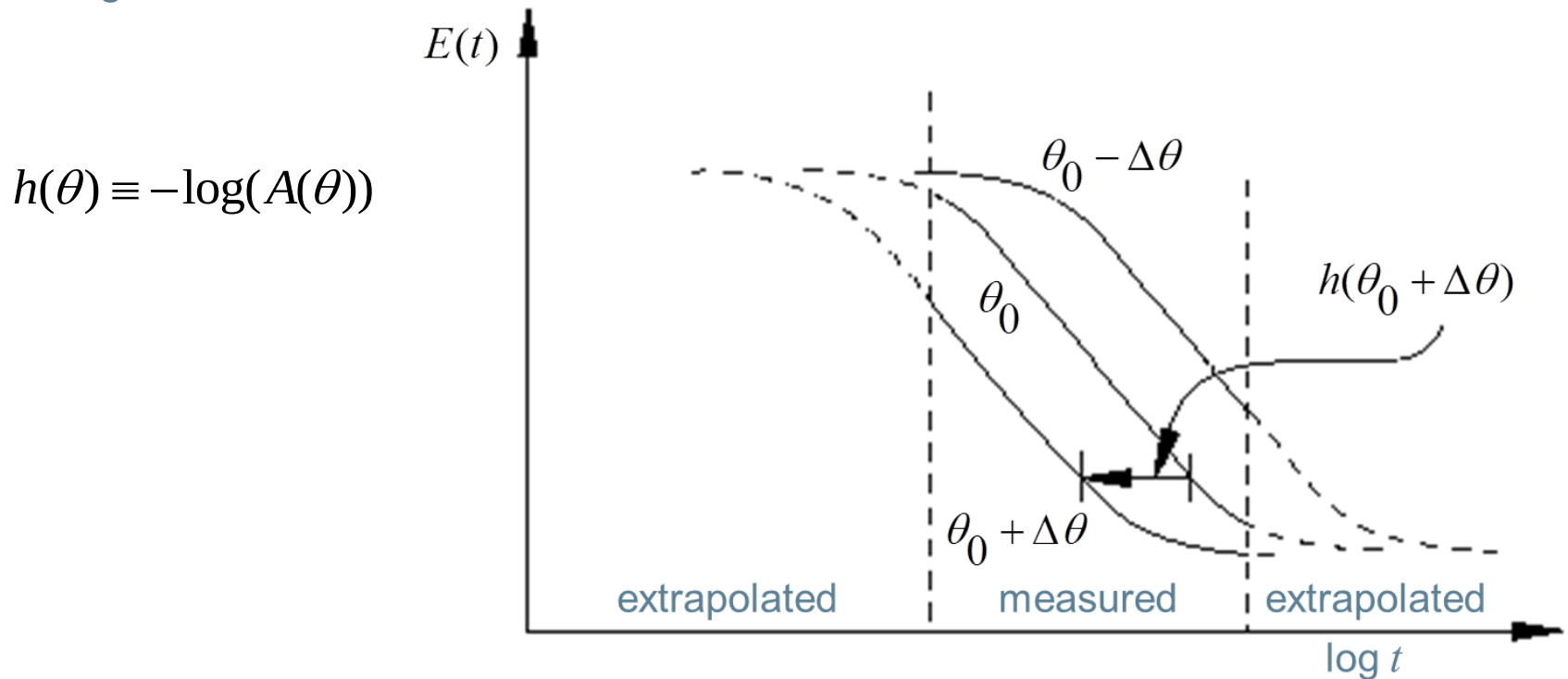
- ▶ For elastomers and plastics, we have the functionality around time-temperature superposition and *TRS
- ▶ Discussed in the *Modeling Rubber and Viscoelasticity with Abaqus* course
- ▶ Experiments showed this time-temperature relationship:

Modulus at 10 minutes at 20°C
= modulus at 10×0.03 minutes at 70°C.

Called *thermo-rheologically simple*



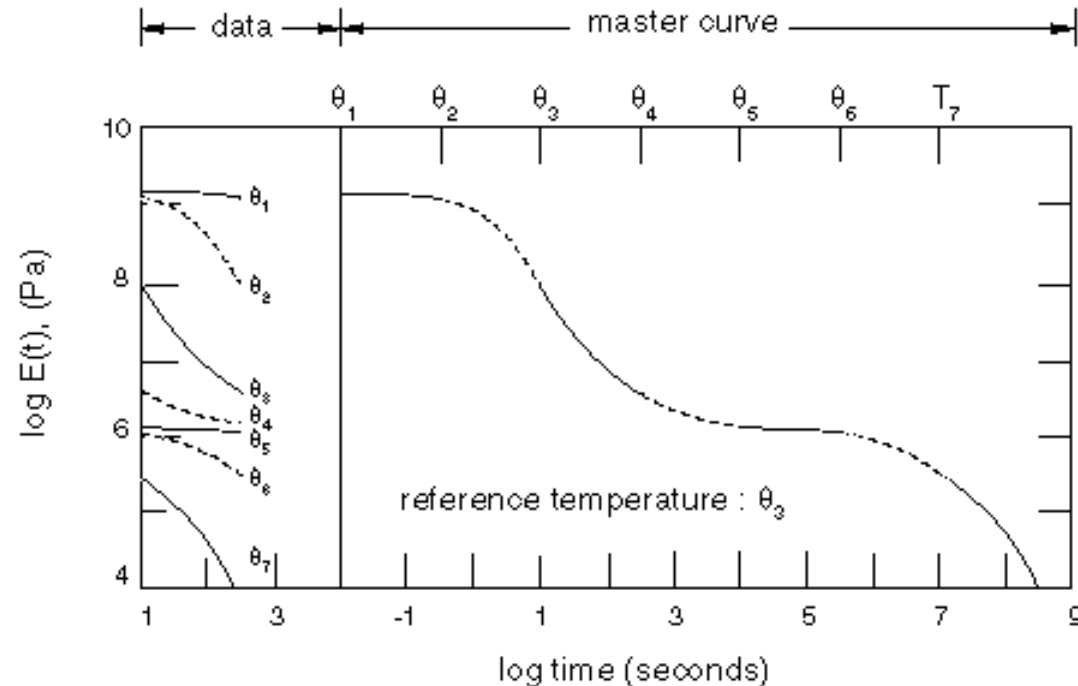
- This time-temperature correspondence is often represented by the logarithmic time shift:



- As shown in this figure, when E is plotted for various temperatures against the log of t , then $h(\theta)$ is the horizontal shift of the curve at temperature θ from the curve at the reference temperature θ_0

Thermo-Mechanical Material Models

- By carrying out relaxation experiments over a wide enough range of temperatures, a complete relaxation curve spanning many decades in time can be obtained



- See Mercier *et al.* for a detailed example of determining the shift function
- Or: <http://home.engineering.iastate.edu/~mkessler/Flash%20Modules/SuperpositionWLF.swf>

- ▶ Abaqus supports the following forms of the shift function $h(\theta)$
 - ▣ Williams-Landel-Ferry (WLF)
 - ▣ Arrhenius form
 - ▣ User-defined forms, UTRS

► Williams-Landel-Ferry (WLF)

- The shift functions for materials about a reference temperature commonly fit the Williams-Landel-Ferry (WLF) form

$$h(\theta) = -\log_{10}(A) = \frac{C_1(\theta - \theta_0)}{C_2 + (\theta - \theta_0)}.$$

- C_1 and C_2 are material constants at the reference temperature θ_0 .
- Any convenient θ_0 can be chosen.
- Curve-fit of measured values of $h(\theta)$ is used to calibrate C_1 and C_2 .
- Since relative temperatures are used, any appropriate temperature scale can be chosen.
- The WLF function is the default shift function in Abaqus

Thermo-Mechanical Material Models

- ▶ References for similar material models for Polyvinyl Butyral:
- ▶ Bennison, S. J., A. Jagota, and C. A. Smith, “Fracture of Glass/Poly(vinyl butyral) (Butacite®) Laminates in Biaxial Flexure,” *Journal of the American Ceramic Society*, vol. 82, pp. 1761–1770, 1999
- ▶ Duser, A., A. Jagota, and S. Bennison, “Analysis of Glass/Polyvinyl Butyral Laminates Subjected to Uniform Pressure,” *Journal of Engineering Mechanics*, vol. 125, no. 4, 435–442, 1999
- ▶ P. D’Haene and G. Savineau, “Mechanical properties of laminated safety glass - FEM Study,” *Proceedings of GPD2007 - Glass Performance Day*, Tampere (Finland), June 15–18, 2007
- ▶ M. Froli and L. Lani, “Adhesion, Creep and Relaxation Properties of PVB in Laminated Safety Glass,” *Proceedings of GPD 2011- Glass Performance Days*, Tampere, (Finland), 2011
- ▶ See also www.glassfiles.com

Constructing a Hyper-Prony-TRS Model

Demo



A horizontal banner for the SIMULIA Learning Community. On the left, the text "SIMULIA Learning Community" is displayed in white on a teal background. To the right is an illustration of several stylized human figures in blue and orange, some standing on a circular path. Further right is a green button with the text "Join Now!" in white. On the far right is a black and white QR code.

www.3ds.com/SLC



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Constructing a Hyper-Visco-TRS Model

- ▶ Professor Allan Bower has created a workshop problem as part of solidmechanics.org
- ▶ Let's look at problem 3.6.5:
http://solidmechanics.org/problems/Chapter3_6/Chapter3_6.php
- ▶ This workshop gives tabular data for $G(t)$ testing at 5 different temperatures and asks the student to:
 - Graph the test data
 - Create a master curve
 - Fit a Prony series to the master curve
 - Plot the shift function versus temperature
 - Fit a WLF equation to the shift function, determining C_1 , C_2
 - Create an Abaqus model and run all 5 simulations
 - Compare the Abaqus responses back to the test data

Constructing a Hyper-Visco-TRS Model

- ▶ Open the spreadsheet “**Bower_TRS_Example.xlsx**” and walk through the solution to this workshop problem.
- ▶ Or see the narrated video on the [SIMULIA Learning Community](#)

27 °C		35 °C		45 °C		55 °C		65 °C	
Time	Modulus	Time	Modulus	Time	Modulus	Time	Modulus	Time	Modulus
(Sec)	(GPa)	(Sec)	(GPa)	(Sec)	(GPa)	(Sec)	GPa	(Sec)	GPa
9	23.95	13	18.89	17	8.81	15	5.67	17	4.92
17	23.91	23	18.46	31	8.61	27	5.16	30	4.86
29	23.84	41	18.04	55	8.34	48	5	55	4.76
52	23.72	73	17.36	98	7.91	85	4.97	97	4.59
93	23.72	131	16.27	174	7.27	152	4.95	174	4.32
166	23.51	233	14.67	310	6.46	271	4.92	309	3.92
295	23.16	415	12.63	552	5.67	482	4.86	550	3.41
525	22.61	738	10.61	982	5.16	857	4.76	979	2.92
933	21.79	1312	9.29	1746	5	1524	4.59	1742	2.6
1660	20.73	2334	8.81	3106	4.97	2711	4.32	3097	2.5
2952	19.66	4151	8.61	5523	4.95	4822	3.92	5508	2.5
5250	18.89	7381	8.34	9822	4.92	8574	3.41	8000	2.5
9337	18.46	13126	7.91	17467	4.86	15248	2.92	17500	2.5
16600	18.04								

► Graph of the Test Data

